

AI / Embodied Intelligence / Embedded Systems

Zhang Quanrui

BUPT / Telecommunications Engineering with Management / Class of 2029

zzzqr007@gmail.com
github.com/zzzqr007-g
Study Room

Profile

BUPT undergraduate with a first-year weighted average of 90.85/100 (top 7%). Building research and engineering experience in embodied intelligence, time-series forecasting, and robotic systems through PRIS Lab training and independent experimentation.

Education

BUPT / Telecommunications Engineering with Management — **Class of 2029**

B.Eng. Candidate / Sep 2025 - Jun 2029 (Expected)

Weighted average 90.85/100 / Top 7% / CET-4 530

- **Academics:** First-year weighted academic average of 90.85/100, ranking in the top 7% of the major.
- **Coursework:** Advanced Mathematics I (88), Advanced Mathematics II (89), Linear Algebra (90), Introduction to Computing and Programming (91), Introduction to Computing and Programming Lab (99), and Fundamentals of Electronic Systems (98).
- **English:** College English Test Band 4 (CET-4): 530.

Research Experience

Pattern Recognition and Intelligent Systems Lab (PRIS Lab) — **Ongoing**

Undergraduate Researcher / Advisor: Prof. Gao Sheng

Embodied AI reading / Blueberry-picking robot / System integration

- **Focus:** Develop a connected understanding of agents and embodied intelligence, from model mechanisms to robotic system constraints.
- **Outcome:** Applied the lab training in a blueberry-picking robot competition project, covering environment setup, serial and teleoperation development, vision-control integration, and mobile-platform testing.

Short-term Bike-sharing Demand Forecasting with LSTM — **Experimental report completed**

Independent Experiment, Analysis, and Report

Python / LSTM / 24-hour input / 8-hour forecast

- **Problem:** Forecasted the next eight hours of bike demand from the preceding 24 hours, using chronological training, validation, and test splits.
- **Method:** Processed about 100,000 Shanghai Mobike orders into 713 hourly samples for imputation, normalization, and a two-layer LSTM; later evaluated about one million Shenzhen open-data orders with cyclical, weekend, and peak-hour features.
- **Results and limits:** Initial tuning reduced MAE from 77.82 to 61.45 and RMSE from 105.26 to 91.61. The expanded-data experiment reached an R^2 of roughly 0.3-0.4; sparse spikes and a 28.6% zero-value share were documented as major limitations.

BUPT-Tsinghua Embodied Intelligence Knowledge Base — **Internal knowledge base**

Core Contributor / University Embodied Intelligence Alliance

124 scholars / 122 labs / Structured academic resources

- **Role:** Owned information collection, manual verification, and structured entry for the Core Scholars and Core Labs modules.
- **Scale:** Completed structured records for 124 scholars and 122 laboratories, supporting a searchable and extensible academic resource base.



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Engineering Projects

Embedded Control System for a Blueberry-picking Robot Based on Renesas RA8P1 — **Project received a provincial second prize**

Primary Developer / Embedded, Vision-control Integration, and Delivery

Environment setup / Serial communication / Arm teleoperation / YOLO integration / Hardware testing

- **Contribution:** Handled most of the engineering implementation, including environment setup, serial communication, robotic-arm teleoperation code, and iterative hardware debugging.
- **Vision and control:** Integrated an existing YOLO detector, converted detections into arm-rotation data and grasp commands, and collected data through recognition and grasp trials; model training was not completed at this stage.
- **System and delivery:** Mounted the arm on a mobile chassis, tested the target-detection, control-data, and grasping pipeline, and prepared the project proposal and presentation deck.

NaviGoose Assistive Sensing System for the Visually Impaired — **Nestling Plan project completed**

AI Scene-recognition Design / Advisor: Assoc. Prof. Zhang Yang

LiDAR / 77GHz mmWave / Edge-cloud architecture

- **Contribution:** In a five-person team, owned AI scene-recognition planning and architecture analysis, and independently prepared the technical proposal, competitor research, and defense materials.
- **Approach:** Compared edge-only, cloud-only, and hybrid architectures, proposing edge-side real-time avoidance with cloud-side 3D reconstruction and path planning.

HarmonyOS Embedded App Development / Calculator — **Completed**

International School Starlight Project

HarmonyOS / Embedded application / Frontend interaction

- **Practice:** Designed and developed a basic calculator application, learning the HarmonyOS workflow, UI organization, and embedded-app debugging process.

Study Room / Immersive Web Study Space — **Live**

Independent Product and Frontend Development / GitHub Pages

React 19 / TypeScript / Vite / Zustand / PWA

- **Positioning:** Combines daily tasks, focus timers, ambient audio, study statistics, and completion feedback in one low-distraction interface.

Machine Learning Portfolio

Machine Learning and Deep Learning Notes: Mechanism-first notes with formulas, runnable practice, and review checks.

ML fundamentals, Neural networks and deep learning, Self-Attention, Transformer, BERT, Practice and review / Curated public edition in progress

Skills and Tools

Programming and Data	C/C++, Python; NumPy, Pandas, PyTorch, and Scikit-learn.
Models and Algorithms	LSTM, Transformers, and attention mechanisms; continuing study of VLA models and embodied intelligence.
Embedded Systems	Renesas RA8P1, e2 studio, serial communication, robotic-arm teleoperation, YOLO inference integration, and hardware testing.
Research and Development Practice	Literature search, paper reading, and experiment logging; Zotero, Anaconda, Markdown, plus AI-assisted programming, debugging, and technical documentation.

Selected Awards and Activities

- CSP-J Olympiad in Informatics / Provincial Second Prize
- Lab4AI × BUPT AI-powered Academic Writing Workshop / Completion Certificate
- BUPT 2025 Nestling Plan / Contributor to a Completed Project
- International School Freshman Debate Tournament / Top 6